



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded

ISE-I (Semester-I) 2025-26

Class: First Year M. Tech.

Branch: All branches

Course: Open Elective: Operations Research

Course Code: OEC-803

Date: 17-Oct-2025

Time: 16:30 to 17:15

Max. Marks: 20

Q. No.	Questions	Marks	CO	BT
Q. 1)	Define systems engineering. Differentiate between systems approach and system analysis. How is it related to operations research?	07	1	1
Q. 2)	What is a model? Explain in brief role of systems models and state the different types of system models.	07	3	2
Q. 3)	Explain the concept of system with a neat diagram.	06	4	2

(Page 1 of 2)



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electronics and Telecommunication Engineering
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: M. Tech. (AI) First Year	Subject: Digital Image and Video Processing	Subject Code: PCC-AI-502
Date: 13/10/2025	Time: 16:30 to 17:15	Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	Explain the fundamental steps in digital image processing.	5	CO1	BT1
Q.2	Discuss in detail the process of uniform sampling and Quantization.	5	CO2	BT2
Q.3	Explain luminance and brightness and contrast with respect to human vision?	5	CO2	BT2
Q.4	Write short note on Color image fundamentals.	5	CO1	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand theory and models in Image and Video Processing
CO 2	Interpret and analyze 2D signals in frequency domain through image transforms.
CO 3	Apply quantitative models of image and video processing for various engineering applications.
CO 4	Develop innovative design for practical applications of image and video processing in various fields.